SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

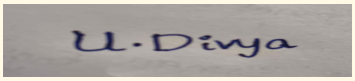
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – EXPERIMENTATION

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – EXPERIMENTATION
Weightage:	30 % of CIA	**Total Marks:**	100 marks
CO2 Statement:	*Compare mobile communication technologies, including multiple access techniques and network evolution, based on performance and applications.*		
Student Name:	U.Divya	**Reg. No.:**	192372277
Section / Batch:	CSE-AI/4th year	**Date:**	26/10/2026

Code of Conduct

I U.Divya/192372277 (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 

Instructions

- This test contains FIVE questions.
- Duration: 60 minutes (homework)

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Annexure A - QUESTIONS

1. Figma-Based Mobile IP Communication Workflow Analysis (BL4 – Analyze)

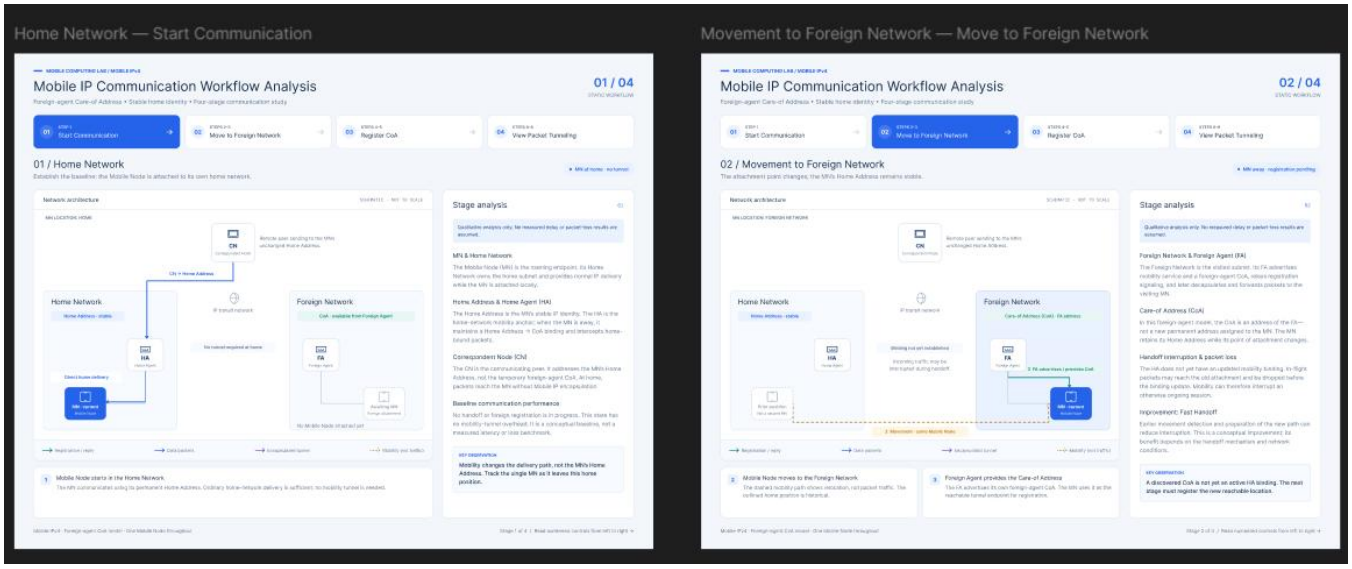
Design an interactive Figma prototype illustrating the complete workflow of Mobile IP communication when a Mobile Node moves from its Home Network to a Foreign Network. Your design should include the Home Agent (HA), Foreign Agent (FA), Correspondent Node (CN), Care-of Address (CoA), Registration Process, and Packet Tunneling.

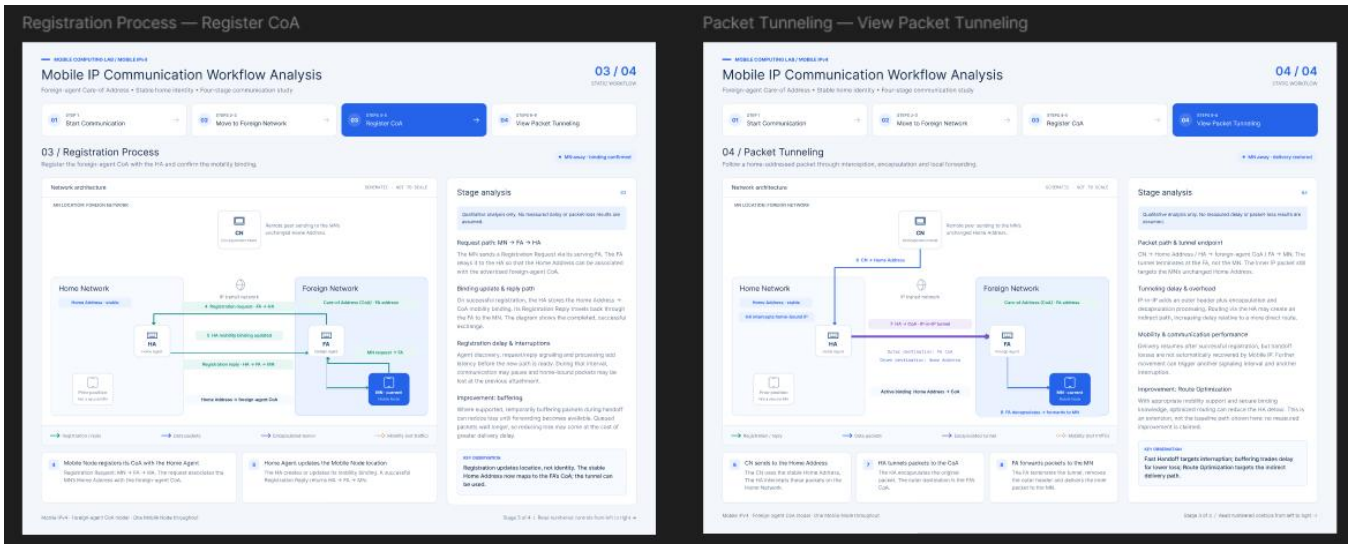
Analyze the communication process by identifying:

- The purpose of each network component.
- Points where packet delay or packet loss may occur.
- The effect of node mobility on communication performance.
- Possible improvements to reduce latency.

Deliverables:

- Figma (.fig) design
- GitHub repository containing exported images/PDF and documentation (README).





2. Comparative Network Architecture Design Using Figma (BL4 – Analyze)

Using Figma or draw.io, create a comparative architectural diagram showing both a Cellular Wireless Network and a Mobile Ad Hoc Network (MANET) operating in the same smart city environment.

Analyze the architectures by comparing:

- Infrastructure dependency.
- Routing mechanisms.
- Scalability.
- Fault tolerance.
- Mobility support.
- Suitable application scenarios.

Highlight the advantages and limitations of each architecture through annotations in your design.

Deliverables:

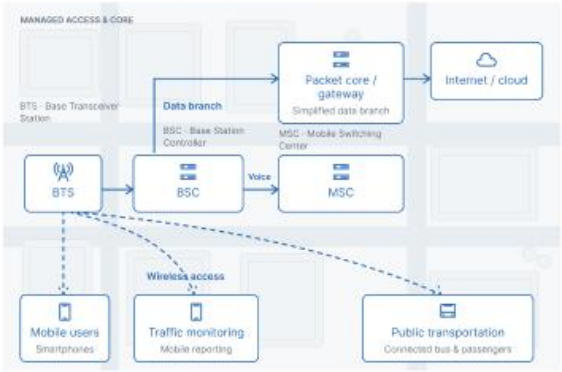
- Editable design file.
- GitHub repository with screenshots, comparison report, and design assets.

Comparative Network Architecture Design: Cellular Wireless Network vs MANET

Two ways to connect a smart city: managed infrastructure versus a self-organizing wireless mesh.

01 Cellular Wireless Network

Simplified GSM circuit-switched model + data branch



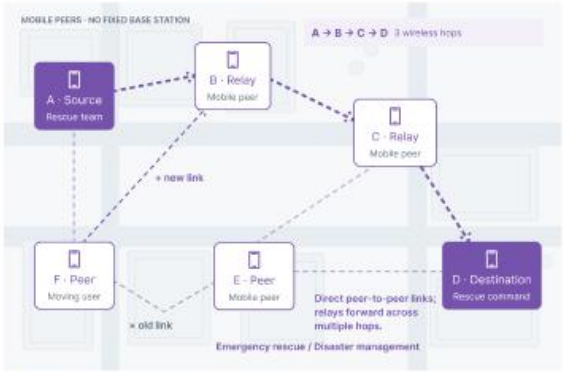
Voice follows BTS → BSC → MSC. Data uses a separate, simplified packet-core/gateway branch; the MSC is not an Internet packet gateway.

+ ADVANTAGES Centralized management · Wide coverage · Reliable infrastructure

− LIMITATIONS Infrastructure dependency · Higher deployment cost

02 Mobile Ad Hoc Network (MANET)

Distributed routing · Mobile wireless nodes



Peers discover and maintain routes cooperatively (e.g. AODV). Movement may break links and trigger rediscovery. Dashed old/new links are a static topology-change example.

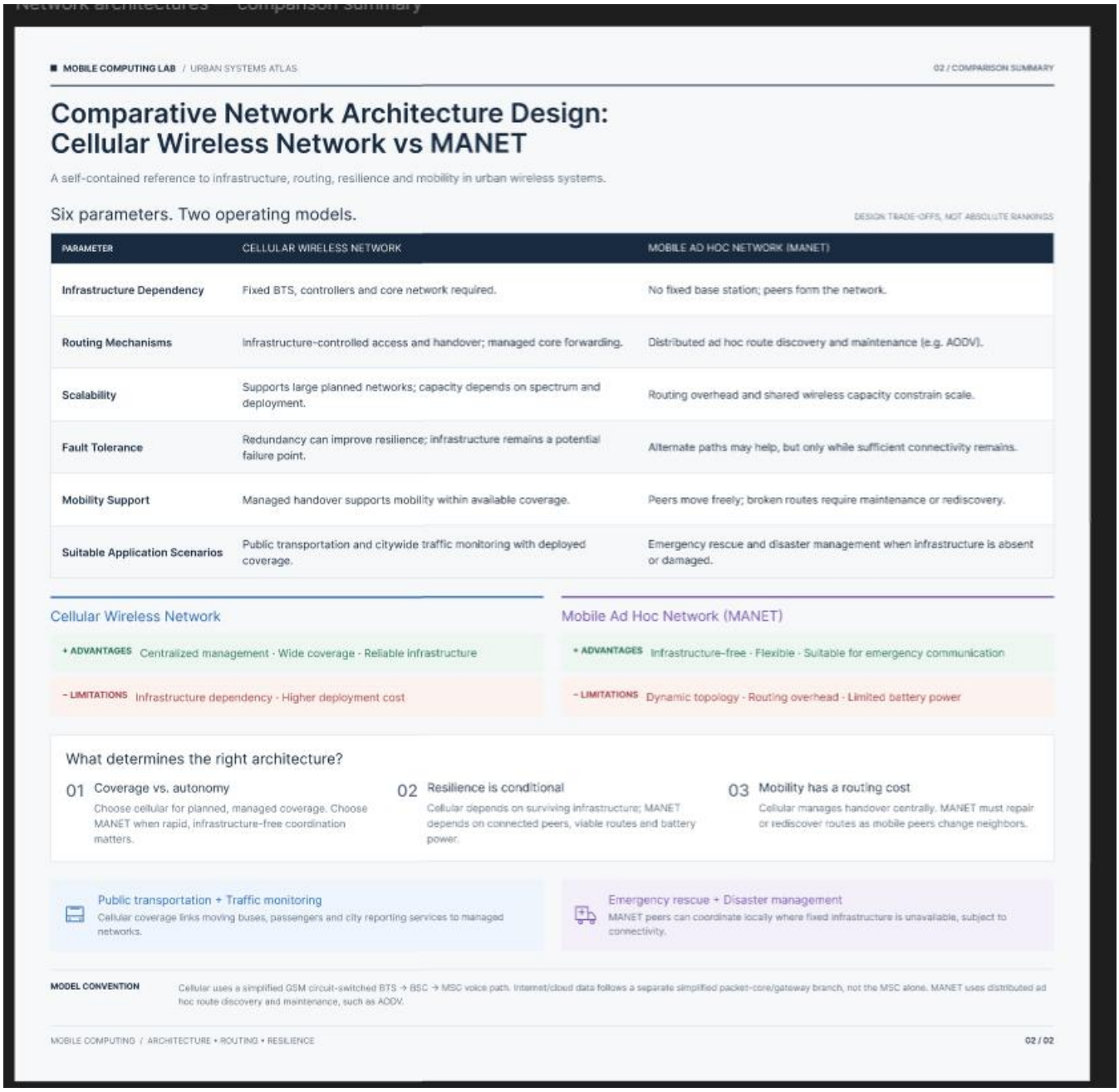
+ ADVANTAGES Infrastructure-free · Flexible · Suitable for emergency communication

− LIMITATIONS Dynamic topology · Routing overhead · Limited battery power

Architecture at a glance

SIX PARAMETERS / QUALIFIED COMPARISON

PARAMETER	CELLULAR WIRELESS NETWORK	MOBILE AD HOC NETWORK (MANET)
Infrastructure Dependency	Fixed BTS, controller and core network	Infrastructure-free; peer-organized
Routing Mechanisms	Infrastructure-controlled access & handover	Distributed discovery & maintenance (AODV)
Scalability	Large planned networks; capacity engineered	Scale constrained by routing overhead
Fault Tolerance	Redundancy helps; infrastructure failure points	Alternate paths depend on connectivity
Mobility Support	Managed handover within network coverage	Movement may break routes; rediscovery needed
Suitable Application Scenarios	Public transportation · Traffic monitoring	Emergency rescue · Disaster management



3. Routing Protocol Visualization and Analysis (BL4 – Analyze)

Develop an interactive visualization using Figma, Mermaid.js, D3.js, or any visualization framework illustrating the operation of one Proactive Routing Protocol and one Reactive Routing Protocol in a mobile wireless network.

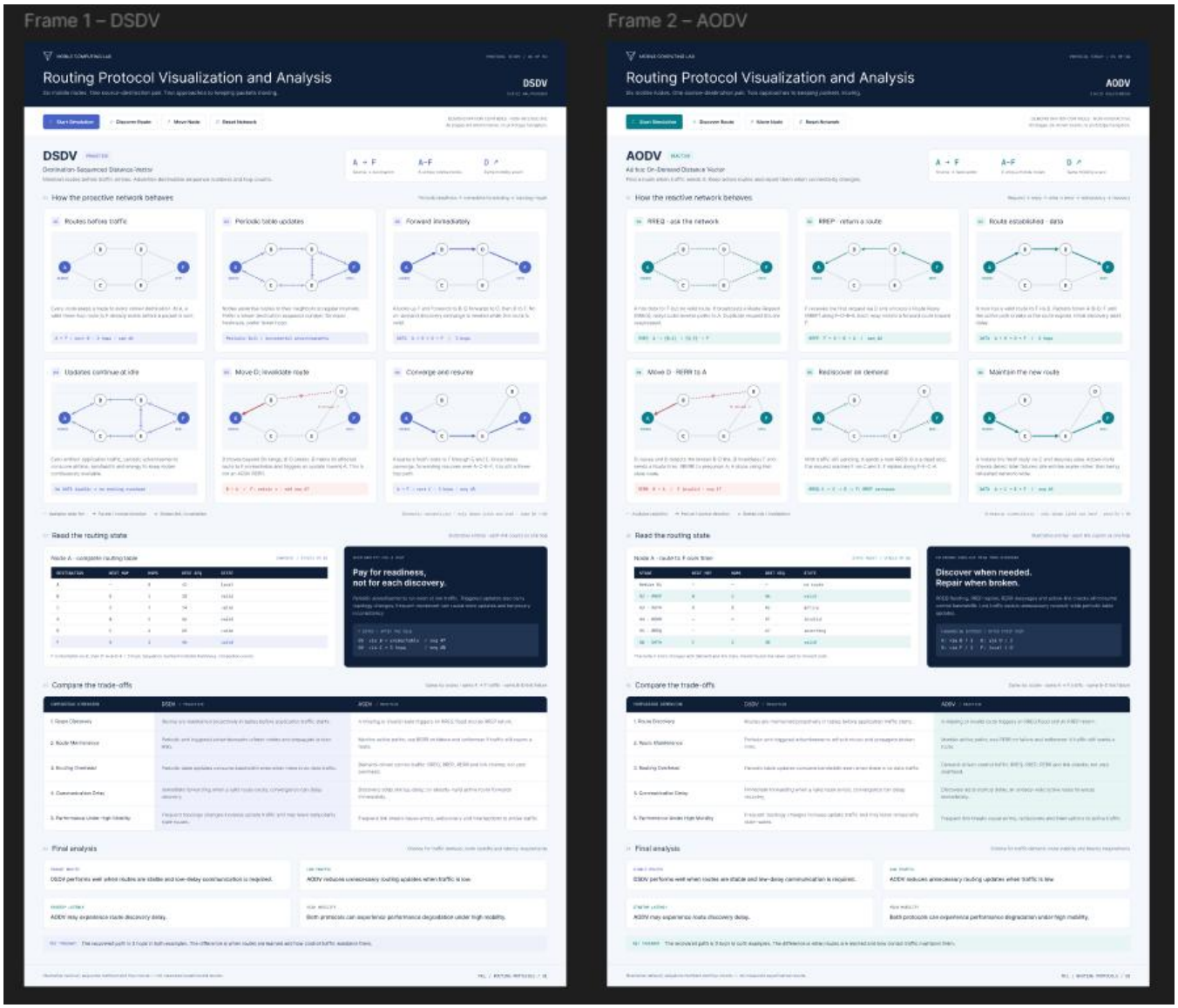
Analyze:

- Route discovery process.
- Route maintenance mechanism.
- Routing overhead.
- Delay in establishing communication.
- Performance under high node mobility.

Include a comparative summary explaining which protocol performs better under different mobility conditions.

Deliverables:

- Source code/design files.
- GitHub repository with implementation, screenshots, and analysis report.



4. Mobile IP Application Design for Smart Healthcare (BL4 – Analyze)

Design a Smart Healthcare Mobile IP system using Figma that enables doctors, ambulances, hospitals, and patients to remain connected while moving across different wireless networks.

Analyze:

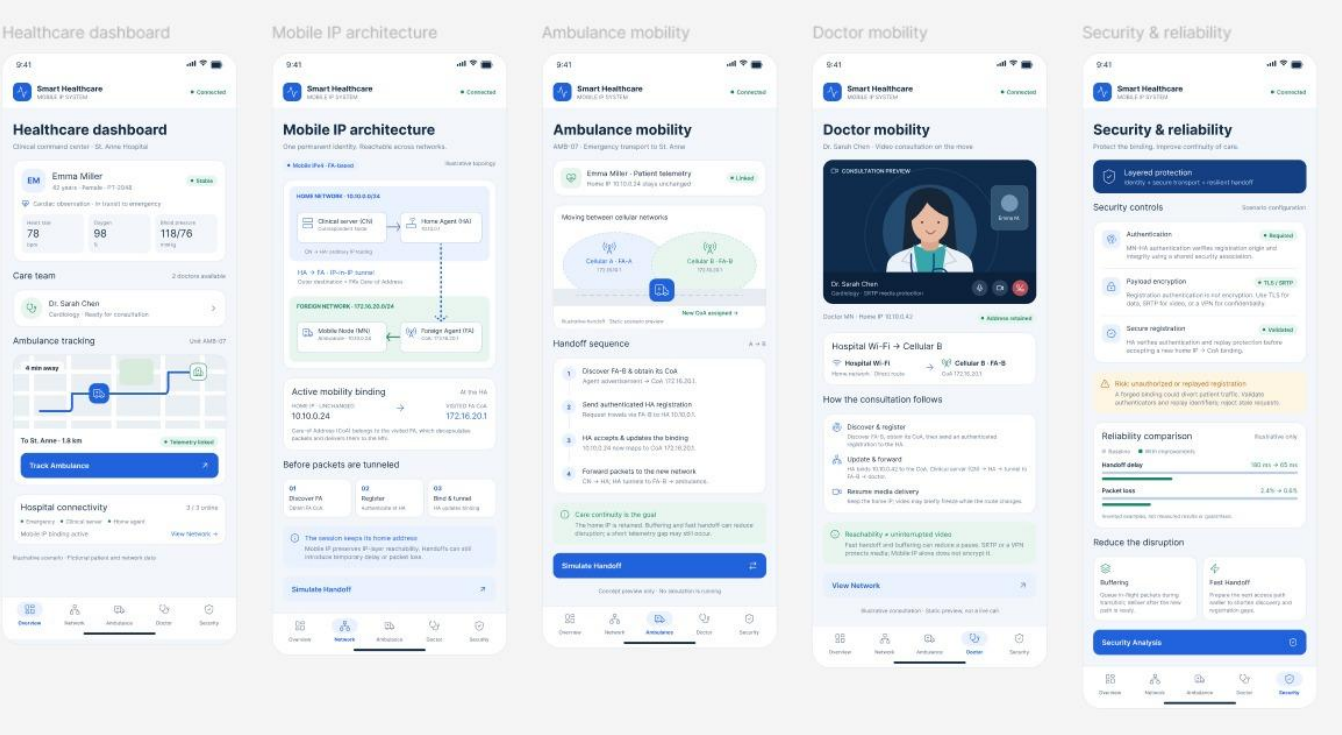
- How Mobile IP maintains uninterrupted communication.
- Challenges during handoff between networks.

- Security vulnerabilities during mobility.
- Possible design improvements to increase reliability.

Provide navigation between screens demonstrating different mobility scenarios.

Deliverables:

- Interactive prototype.
- GitHub repository containing design files, exported prototype, and documentation.



5. Wireless Network Decision Dashboard Using GitHub Project (BL4 – Analyze)

Create a web-based dashboard (using HTML/CSS/JavaScript, React, or another programming framework) or an interactive Figma dashboard that helps network engineers determine whether a given scenario should use Mobile IP, Cellular Networks, or Ad Hoc Networks.

The dashboard should analyze multiple parameters such as:

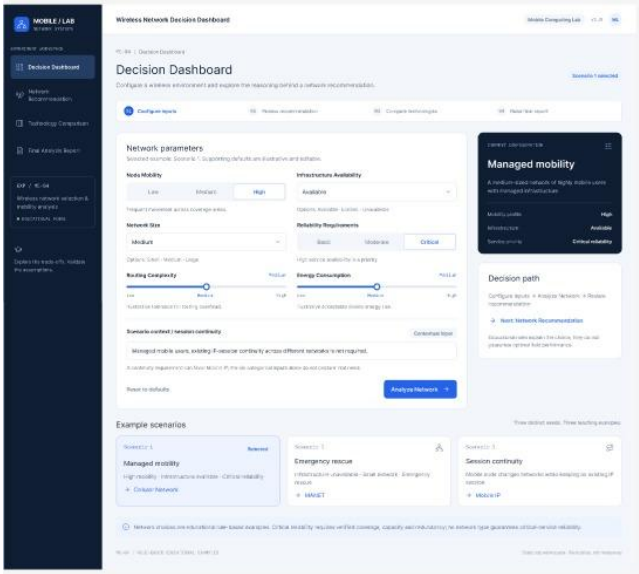
- Node mobility.
- Infrastructure availability.
- Network size.
- Routing complexity.
- Energy consumption.
- Reliability requirements.

The system should justify its recommendation based on the analyzed parameters.

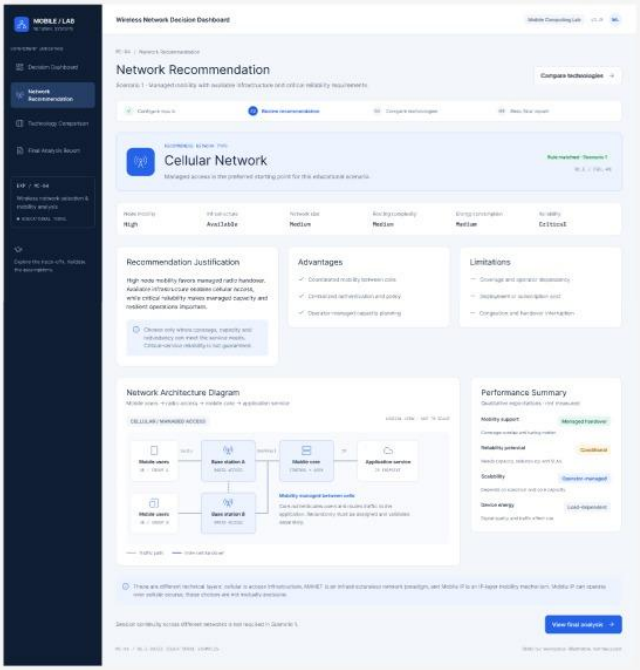
Deliverables:

- Source code or Figma project.
- GitHub repository with complete code/design, README, screenshots, and analysis report.

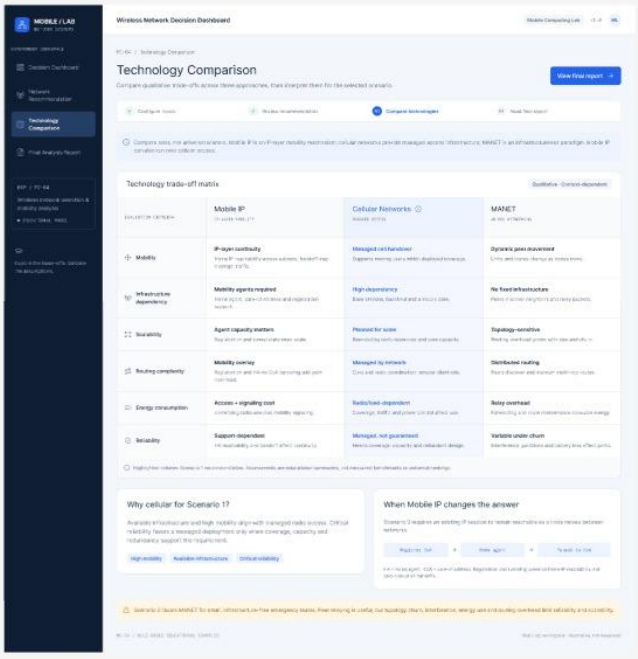
Decision Dashboard



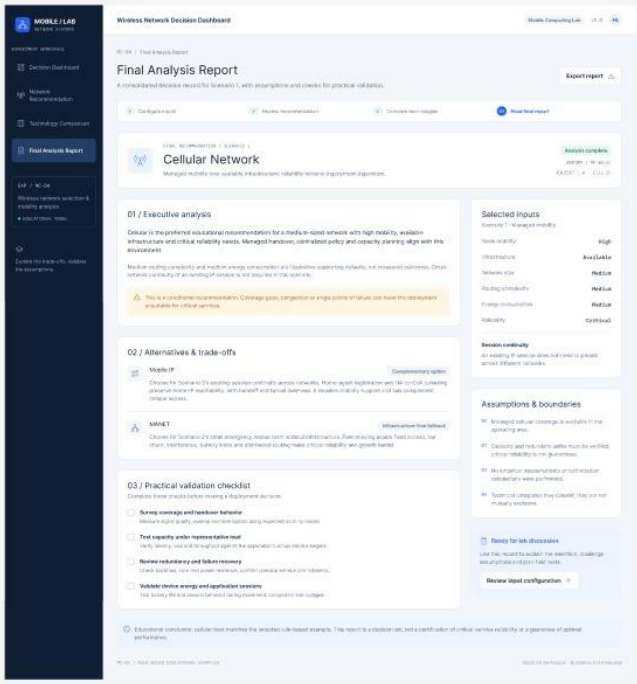
Network Recommendation



Technology Comparison



Final Analysis Report



Criterion	Marks
Design Quality (Figma/Programming Tool)	20
Technical Analysis	30
Correctness of Mobile Networking Concepts	20
GitHub Repository Organization & Documentation	15
Creativity and Visualization	15
Total	**100 Marks**